

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0699

Pana North 138 kV
September 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0699. The Interconnection Customer proposes a 83.1 MW solar photovoltaic facility interconnecting at the Pana North 138 kV substation in the West Virginia area. The request was received on November 07, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 240 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$84,362,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HUSTON-GRANA 1907-1909 138 kV	FISHKILL PLAINS-DILLARD-ORANGE 345 kV Line	139.9	285	22.1
GRANA 1907-1909-PANA NORTH 138 kV line	FISHKILL PLAINS-DILLARD-ORANGE 345 kV	138.5	294	22.1
MIL-HI-SEMAHMOO 138 kV	FISHKILL PLAINS-DILLARD-ORANGE 345 kV	127.0	282	8.3
MARQUETTE ENERGY CTR-BOOSTER SUBSTATION 138 kV	DILLARD-ORANGE-FISHKILL PLAINS 345 kV	123.2	223	11.9
MARQUETTE ENERGY CTR-HUSTON 138 kV	DILLARD-ORANGE-FISHKILL PLAINS 345 kV L	122.7	228	11.9
RIVERTON 115 SUBSTATION-BORDER STEEL 138 kV line	MARQUETTE ENERGY CENTER-HUSTON 138 kV	121.2	208	9.8
RIVERTON 115-SPENCER GAS TURBINE SUBSTATION 138 kV	MARQUETTE ENERGY CTR-HUSTON 138 kV	121.1	211	9.8
CEDAR CREST SUBSTATION-SPENCER GAS TURBINE 138 kV	MARQUETTE ENERGY CTR-HUSTON 138 kV	120.7	212	9.8
MILLBRAE-CHESTER GENERATING STATION 138 kV L	HALSEY-BORDER STEEL 138 kV	119.7	212	5.3
CEDAR CREST-LOUISIANA PACIFIC 138 kV line	MARQUETTE ENERGY CTR-HUSTON 138 kV	119.5	217	9.8
LOUISIANA PACIFIC-GOOSPORT 138 kV	MARQUETTE ENERGY CENTER-HUSTON 138 kV L	116.1	225	9.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CAROLANDS-BOOSTER 138 kV L	LOUISIANA PACIFIC-GOOSPORT 138 kV	113.8	210	9.3
LODI-HALSEY 138 kV	MILLBRAE-CHESTER GENERATING STA 138 kV	110.0	206	5.3
HALSEY-BORDER STEEL 138 kV	MILLBRAE-CHESTER GENERATING STATION 138 kV	109.7	211	5.3
CAROLANDS SUBSTATION-NOVATO 138 kV Line	LOUISIANA PACIFIC-GOOSPORT 138 kV	107.0	220	9.3
LODI-MORGANTOWN ENERGY FACILITY 138 kV L	MILLBRAE-CHESTER GENERATING STATION 138 kV	103.6	194	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HUSTON-GRANA 1907-1909 138 kV	Rebuild line with higher-capacity conductor	\$10,717,000	15% (\$1,597,000)
GRANA 1907-1909-PANA N 138 kV	Rebuild line with higher-capacity conductor	\$5,987,000	15% (\$892,000)
SEMAHMOO-MILE-HI 138 kV	Rebuild line with higher-capacity conductor	\$2,702,000	8% (\$219,000)
MARQUETTE ENERGY CTR-BOOSTER 138 kV	Rebuild line with higher-capacity conductor	\$13,076,000	100% (\$13,076,000)
MARQUETTE ENERGY CTR-HUSTON 138 kV	Rebuild line with higher-capacity conductor	\$11,976,000	100% (\$11,976,000)
RIVERTON 115 SUBSTATION-BORDER STEEL 138 kV line	Rebuild line with higher-capacity conductor	\$5,744,000	100% (\$5,744,000)
RIVERTON 115-SPENCER GAS TURBINE 138 kV	Rebuild line with higher-capacity conductor	\$11,677,000	100% (\$11,677,000)
SPENCER GAS TURBINE-CEDAR CREST 138 kV	Rebuild line with higher-capacity conductor	\$4,217,000	100% (\$4,217,000)
MILLBRAE-CHESTER GENERATING STA 138 kV Line	Rebuild line with higher-capacity conductor	\$10,824,000	32% (\$3,413,000)
LOUISIANA PACIFIC-CEDAR CREST 138 kV line	Rebuild line with higher-capacity conductor	\$8,243,000	100% (\$8,243,000)
LOUISIANA PACIFIC-GOOSPORT 138 kV	Rebuild line with higher-capacity conductor	\$8,460,000	100% (\$8,460,000)
CAROLANDS-BOOSTER 138 kV	Rebuild line with higher-capacity conductor	\$5,315,000	100% (\$5,315,000)
LODI-HALSEY 138 kV	Rebuild line with higher-capacity conductor	\$14,806,000	27% (\$3,976,000)
HALSEY-BORDER STEEL 138 kV line	Reconductor line	\$2,665,000	27% (\$716,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CAROLANDS SUBSTATION-NOVATO 138 kV Line	Reconductor line	\$4,706,000	100% (\$4,706,000)
LODI-MORGANTOWN ENERGY FACILITY 138 kV L	Reconductor line	\$504,000	27% (\$135,000)
	Total allocated to project		\$84,362,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.